CULC Solar Panel Installation

Energy Efficiency VIP 2601

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Executive Summary

As a part of Georgia Tech's Climate Action Plan, there is an amplified push in electrifying buildings on campus to reduce scope 1 emissions, increase renewable energy, and expand bench marking for energy conservation measures. One component of these goals is to increase campus visibility to these missions and one effective way to enable students to join the conservation by having a solar powered workstation that enhances student learning capacity. Within this document is the proposal and rationale for a solar powered table workstation to be placed on the rooftop gardens of the Clough Undergraduate Learning Commons. The Sunbolt Adventurer 400 solar panel system will act as the shading umbrella of a picnic bench style workspace. The 100Ah battery system has been confirmed to meet the varying student electricity demands for standard occupancy of 1-4 students from 12-5pm. Maintenance cost are expected to be low, ~$10-30 per year even with expected seasonal losses and the entire system itself is expected to have a payback period of 21 years.

Introduction

With Georgia Tech looking to expand its efficiency, solar panels are a great start in the right direction. With Atlanta being in the southeast, the sun is plentiful and it is an overlooked source of energy that is not being used to its fullest potential. The Clough Undergraduate Learning Commons rooftop garden is a heavily occupied green area that many students utilize to study. Installing a solar panel on the rooftop would add to the green effect of the space and allow for constant power even during a power outage. With the panel obtaining its main source of energy from the s un, there will be an additional battery to save the unused energy for a later time.

With our solar panel location being on a roof, there is less obstruction that needs to be overcame. The Clough Undergraduate Learning Commons has some obstruction due to the main roof with respect to the garden part, but it will not be an issue as the garden is to the west side of campus. With CULC having a high occupancy around mid to late afternoon, the sun is already on the west side, and the power output can be maximized for that time of day (approximately 12-5PM). The garden is closed at 5, so our angle can be better adjusted to those specific hours as opposed to a wider time range where some energy could be lost. With Atlanta's coordinates, the average ideal angle to maximize power is roughly 29 degrees to the south. With the CULC garden being outdoors, it is less likely that students will occupy that space in the winter months. By narrowing down our constraints to be March- October from 12-5PM everyday, our angle can be lessened and power out put for that time of the day can be maximized. After calculating the average angles of the given months, the final angle to the south was determined to be 23 degrees (see Figure 1 Below). Summer hours are another factor to consider. Campus will be less occupied and the demand for energy will decrease. Therefore the angle can be on the higher side

of the average as students will occupy the garden rooftop during the spring and fall seasons where the angle needs to be greater.

The prepared solar panel array can be modeled using various equations to account for the incident radiation projected from the sun while accounting for seasonal variations in the sun's path. Due to the scalable nature of these table-shade solar installations, the ease of replication with calculations was a consideration, and while creating a calculator was one option, the readily existing PVWatts calculator by the National Laboratories of the Rockies utilizes the same calculations along with additional weather variables from 20 years of data. Using 33.77, -84.38 for our latitude and longitude coordinates with a .4 kW DC system size, standard module type, fixed open rack, 14.08% system losses (reasonable), 23 degrees of tilt, and 180 degrees azimuth (facing south), yields a reasonable power output estimate of 568 kwh/year, which would yield 8520 kWh over a 15 year period. Varying the system losses from zero to 55.71% (assuming 50% from shading losses) provides an output range of the extremes at 657 kwh/year to 291 kwh/year.

Standard calculations: PVWatts Calculator - VIP Spring 2026 Standard Culc calculation.pdf

With the recommended 1.2 kWh battery storage for the unit, the power generation would be more than capable of meeting daily energy demands any month of the year (36 kWh at lowest in December at 14.08% system losses), with the worst case only hindering the winter months when it is already too cold for students to work outside (18 kWh for December at 55.71% system losses). Thus, this configuration would be capable of meeting typical demand needs when at capacity, such as charging a cell phone at 10W up to 120 times or a 50W Laptop for 24 hours of usage. Additional considerations should be made to mitigate the effects of low-demand seasons, particularly in the summer and winter periods where few students utilize the rooftop space. For

these times, it would be ideal to curtail the energy generated as to not put additional risk to the system. Mitigating battery degradation could be done with built in demand loads such as 30W canopy fans which could run continuously for 40 hours on the battery capacity alone, or use LED lighting, which could be run indefinitely coupled with generation and storage though with less utility for students accounting for the 5pm rooftop closure. This would also be useful with a data-tracking unit to monitor generation and demand fluctuations to reference for future iterations of this project.

The proposed solar panels have a capacity of 400 W (0.4 kW) and are expected to generate approximately 584 kWh annually under typical Atlanta weather conditions. This estimate assumes an average of 5 peak sun-hours per day and an operating efficiency of 80%. Using a campus average electricity cost of $0.12 per kWh, this corresponds to approximately $70 per year in avoided electricity costs. Over a 15-year period, the system is expected to generate approximately $1,050 in total savings. Given an estimated total cost of $1,460 for materials and mounting, the payback period is approximately 21 years ($1,460 ÷ $70 ≈ 20.9). Therefore, the system does not fully recover its cost within the 15-year analysis period. In order to return a full return on investment within 15 years, the required electricity rate would need to be approximately $0.17 per kWh, which is higher than current campus electricity rates. Additional costs include maintenance, such as cleaning and potential component degradation over time. Estimated maintenance costs are relatively low, at approximately $10–$30 per year. However, neglecting maintenance may reduce system efficiency due to dust accumulation or component wear. For example, dirty panels can reduce energy production by approximately 10–15%. Additionally, Although the system is small in scale, it generates electricity during peak daytime hours when demand is highest, contributing to demand offset from the grid.

Approximately 75% of the system's energy production occurs during peak-demand periods, allowing it to offset about 438 kWh of peak-period energy annually, which corresponds to roughly $66 in peak electricity value. The solar panel also contributes to the grid resilience by providing localized power. Coupled with the battery, the panel can continue to supply energy, although limited, during outages

Site Justification for Solar Panel

Proposed site: Rooftop of the CULC (Clough Undergraduate Learning Commons)

Coordinates

Latitude: 33° 46' 27.69" N

Longitude: -84° 23' 47.27" W

Figure 1 Original solar powered work station

The Clough Undergraduate Learning Commons is a 5 story 222,000 square foot mixed use academic space which began construction in 2010. It serves as one of the three connected student productivity hubs in a central region of campus. The 18,000 square foot rooftop garden of the CULC, one of the few full-sun outdoor productivity spaces for students and staff, is open M-F from 8am-5pm.

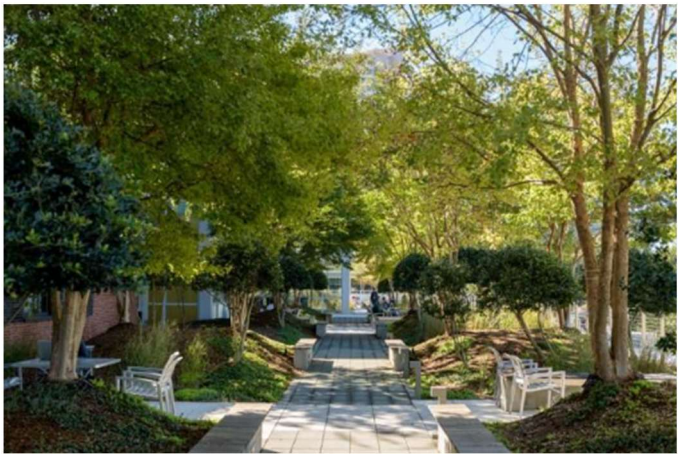
Figure 2 Overhead diagram of the roof of the CULC

It has been observed on several occasions by students that the outlets currently available on the benches alongside the garden banks are unable to be used. This is likely due to water damage judging by their rusted appearance. The current placement of the outlets makes them difficult to access from available seating. Students who need an outlet work with their laptops or notebooks on their lap.

The proposed solar workstation aims to address these student needs. A comfortable outdoor workspace that meets energy needs and offers a full spectrum view of campus and of the skyline. Exact temporal occupancy data is not available for the rooftop of the CULC through waitz.io/gatech which has occupancy data available for all other study spaces on campus. Instead, estimates from observed occupancy by frequent visitors of the space will be used. Below are estimates for low and highest occupancy scenarios for laptop and phone charging.

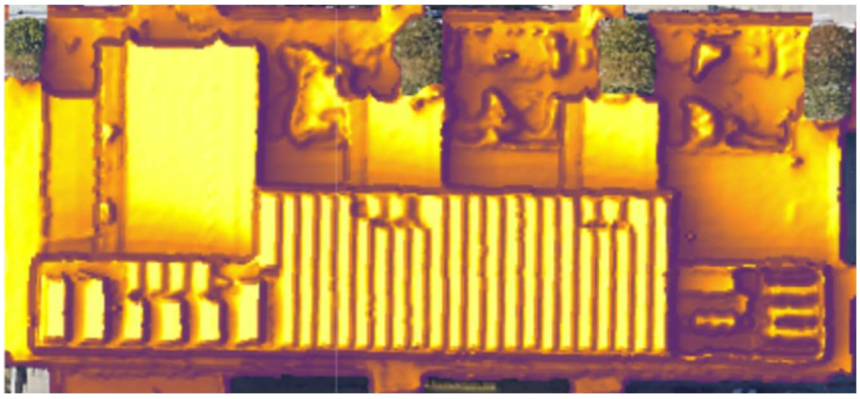
Number of Occupants	Duration of use	Total Energy Consumed
1	2 hours	700Wh
4	2 hours	2800Wh

Figure 3 Shaded regions of the rooftop garden during Summer and early Fall



During the summer, the roof top has plenty of tree shade cover, however during the peak use period between the late Fall and Spring semester academic sessions, there is no reprieve from the pressing sun as the leaves wither away.

Figure 4 Snapshot of sunlight on the surface of the rooftop garden and the roof of the CULC. The bright vertical lines are current solar panels arrays.



The only other solar powered work station on campus is a work table near the LOVE manufacturing building that is seldom used by students. It requires more “work” (look up the word for having to go out of you way for something) by student in order to access it. By

choosing a high traffic, high demand area for the new solar roof, it is anticipated to receive ample use. Given that there is an existing structure with this design, it makes it easy to replicate and optimize for current student need

Though the Google sunroof project can provide good estimates of solar roof capacity, it has its limitations because much of the data has not been updated since 2018 and based on a case study from Ecowatch, underestimates the cost by more than 50% in some cases (Ecowatch, 2024). Instead, suncalc.org was use to determine the solar constant

Description of Solar Panel Characteristics

The Sunbolt Adventurer 400 is a classified as a high efficiency crystalline silicone PVWatts Standard module type. Its accompanying 100Ah battery can be charged from 50% depth of charge (amount of discharged energy from a battery) within 3 hours. The 100Ah is enough to sustain is a low to mid range scenario of occupancy of our solar work station.

Appendix:

Year	Energy Produ	Annual Savin	Cumulative S	Net ROI ($)
1	584	70	70	-1390
2	584	70	140	-1320
3	584	70	210	-1250
4	584	70	280	-1180
5	584	70	350	-1110
6	584	70	420	-1040
7	584	70	490	-970
8	584	70	560	-900
9	584	70	630	-830
10	584	70	700	-760
11	584	70	770	-690
12	584	70	840	-620
13	584	70	910	-550
14	584	70	980	-480
15	584	70	1050	-410

Figure 1 Energy production and savings over the next 15 years

Monthly South-Facing Angle Data

Month	Angle (°)
March	23
April	17
May	13
June	14
July	20
August	27
September	32
October	36

Average	
Angle (°)	**22.8**

Figure 2 Average angle of the optimal azimuths per month

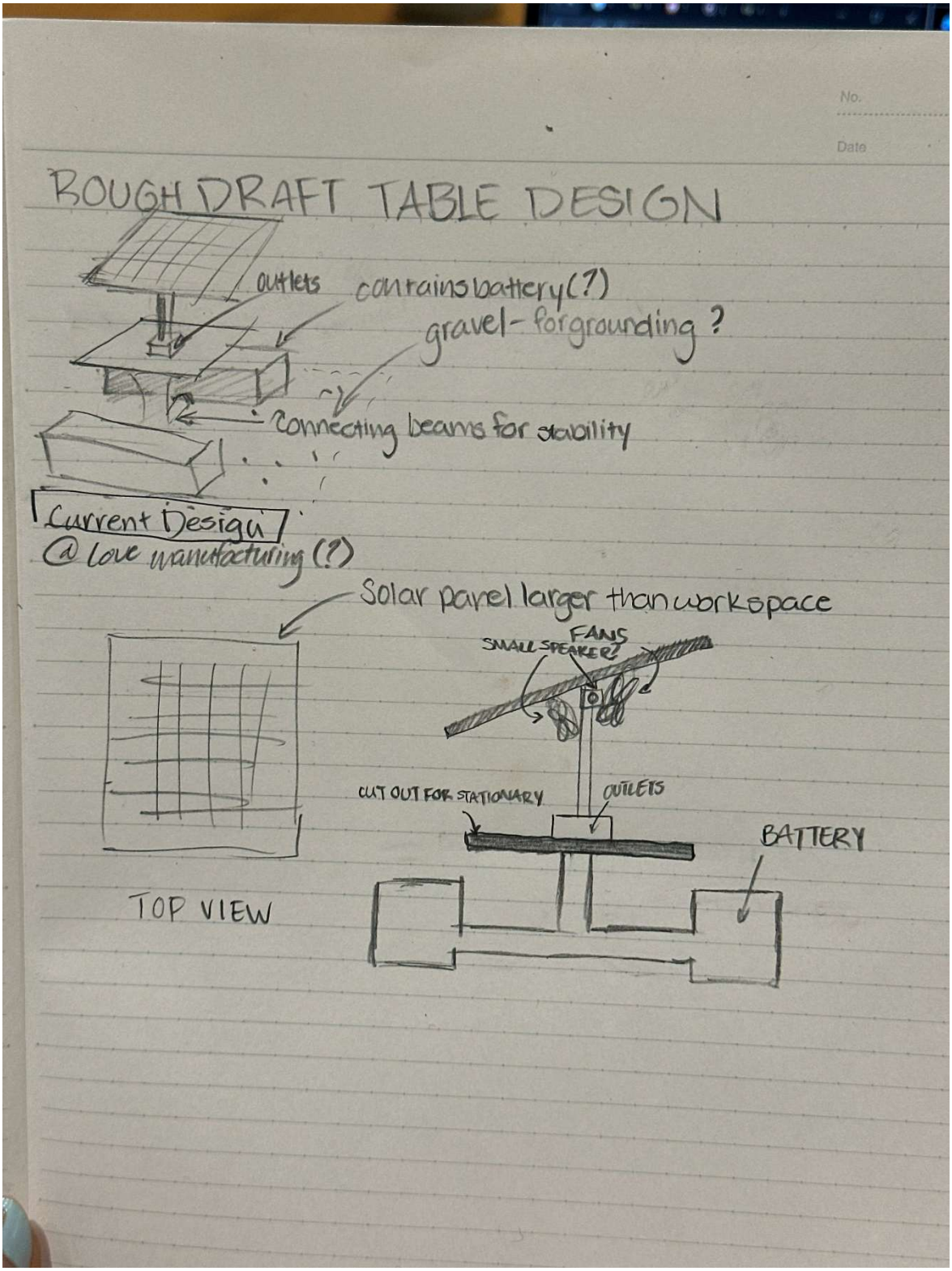


Figure. 3 Sketch of work station with speaker and fan to mitigate battery degradation from idle charge

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